# ETC3410 Applied Econometrics

## Group Assignment 2

## Group 17


**Student 1:** Yilin Huang, 35107731
**Student 2:** Zhiqi Fang, 35954396
**Student 3:** Minghong Xia, 35440562
**Student 4:** Amy Bing, 34940227

# Question 1

## a) Endogeneity

Endogeneity occurs when an explanatory variable is correlated with the error term in the regression model, so that

$$\text{Cov}(x_j, u) \neq 0.$$

This violation of Gauss-Markov assumptions (zero conditional mean assumption) leads to biased and inconsistent OLS estimates.

## b) Instrumental Variables

An instrumental variable is a variable used to address endogeneity by isolating exogenous variation in the endogenous regressor. A valid instrument $z$ must satisfy relevance,

$$\text{Cov}(z, x) \neq 0,$$

and exogeneity,

$$\text{Cov}(z, u) = 0.$$

## c) Overidentification

Overidentification occurs when the number of instrumental variables exceeds the number of endogenous explanatory variables. In this case, there are extra instruments, allowing us to test the validity of the overidentifying restrictions.

## d) Hausman's Test for Overidentifying Restrictions

Hausman's test for overidentifying restrictions tests whether the extra instruments are jointly exogenous. The null hypothesis is that all instruments are uncorrelated with the error term, and the test statistic is

$$H = nR^2 \overset{a}{\sim} \chi^2(q),$$

where $q$ is the number of overidentifying restrictions.

## e) Two-Stage Least Squares Estimator

The two-stage least squares estimator is an IV estimator used when one or more regressors are endogenous. In the first stage, the endogenous variable is regressed on the instruments and exogenous variables; in the second stage, the outcome is regressed on the fitted value from the first stage and the exogenous variables.

# Question 2

The simultaneous equations model is

$$support_i = \delta_{10} + \gamma_{12} visits_i + \delta_{11} finc_i + \delta_{12} fremarr_i + \delta_{13} dist_i + u_{1i}, \tag{1}$$
$$visits_i = \delta_{20} + \gamma_{21} support_i + \delta_{21} mremarr_i + \delta_{22} dist_i + u_{2i}. \tag{2}$$

## (a)

To determine identification, we use the order condition and the rank condition. The jointly determined endogenous variables are $support_i$ and $visits_i$, and the exogenous variables in the system are $finc_i$, $fremarr_i$, $dist_i$, and $mremarr_i$. Since there are $G = 2$ equations in the system, each equation must exclude at least $G - 1 = 2 - 1 = 1$ exogenous variable to be identified.

(i) **Support equation.**

In the support equation, the endogenous explanatory variable is $visits_i$. The included exogenous variables are $finc_i$, $fremarr_i$, and $dist_i$, while the excluded exogenous variable is $mremarr_i$. Thus, the number of excluded exogenous variables is 1, which satisfies the order condition since

$$1 = G - 1 = 1.$$

Since all parameters are assumed to be nonzero, $\delta_{21} \neq 0$, so $mremarr_i$ affects $visits_i$. Hence, the rank condition is also satisfied. Therefore, the support equation is just-identified.

(ii) **Visits equation.**

In the visits equation, the endogenous explanatory variable is $support_i$. The included exogenous variables are $mremarr_i$ and $dist_i$, while the excluded exogenous variables are $finc_i$ and $fremarr_i$. Thus, the number of excluded exogenous variables is 2, which satisfies the order condition since

$$2 > G - 1 = 1.$$

Since all parameters are assumed to be nonzero, $\delta_{11} \neq 0$ and $\delta_{12} \neq 0$, so $finc_i$ and $fremarr_i$ affect $support_i$. Hence, the rank condition is also satisfied. Therefore, the visits equation is overidentified.

## (b)

Since both equations are identified, they can be estimated using two-stage least squares. For the support equation, the endogenous explanatory variable is $visits_i$. In the first stage, regress $visits_i$ on all exogenous variables in the system:

$$visits_i = \pi_0 + \pi_1 finc_i + \pi_2 fremarr_i + \pi_3 dist_i + \pi_4 mremarr_i + v_i.$$

This gives the fitted value $\widehat{visits}_i$. In the second stage, regress $support_i$ on $\widehat{visits}_i$ and the exogenous variables included in the support equation:

$$support_i = \delta_{10} + \gamma_{12}\widehat{visits}_i + \delta_{11} finc_i + \delta_{12} fremarr_i + \delta_{13} dist_i + e_i.$$

The coefficient on $\widehat{visits}_i$ is the 2SLS estimate of $\gamma_{12}$.

For the visits equation, the endogenous explanatory variable is $support_i$. In the first stage, regress $support_i$ on all exogenous variables in the system:

$$support_i = \rho_0 + \rho_1 finc_i + \rho_2 fremarr_i + \rho_3 dist_i + \rho_4 mremarr_i + r_i.$$

This gives the fitted value $\widehat{support}_i$. In the second stage, regress $visits_i$ on $\widehat{support}_i$ and the exogenous variables included in the visits equation:

$$visits_i = \delta_{20} + \gamma_{21}\widehat{support}_i + \delta_{21} mremarr_i + \delta_{22} dist_i + q_i.$$

The coefficient on $\widehat{support}_i$ is the 2SLS estimate of $\gamma_{21}$.

## (c)

To test whether $visits_i$ is endogenous in the father's reaction function, we use a Hausman control-function test.

First, estimate the reduced-form equation for $visits_i$:

$$visits_i = \pi_0 + \pi_1 finc_i + \pi_2 fremarr_i + \pi_3 dist_i + \pi_4 mremarr_i + v_i,$$

and obtain the first-stage residual $\widehat{v}_i = visits_i - \widehat{visits}_i$.

Second, add $\widehat{v}_i$ to the original support equation:

$$support_i = \delta_{10} + \gamma_{12} visits_i + \delta_{11} finc_i + \delta_{12} fremarr_i + \delta_{13} dist_i + \alpha\widehat{v}_i + \varepsilon_i.$$

The hypotheses are

$$H_0 : \alpha = 0 \quad \text{against} \quad H_1 : \alpha \neq 0.$$

Under $H_0$, $visits_i$ is exogenous in the support equation. Under $H_1$, $visits_i$ is endogenous.

The test statistic is

$$t = \frac{\widehat{\alpha}}{\mathrm{se}(\widehat{\alpha})} \overset{a}{\sim} N(0,1) \quad \text{under } H_0.$$

At the 5% significance level, reject $H_0$ if $|t| > 1.96$, or equivalently if the $p$-value is less than 0.05.

If $H_0$ is rejected, we conclude that $visits_i$ is endogenous and 2SLS is preferred to OLS. If $H_0$ is not rejected, there is no statistically significant evidence that $visits_i$ is endogenous.

# Question 3

## 3.1 Summary Statistics

Using the definitions of $selectyrs$ and $choiceyrs$, the required counts are as follows.

**(a)** Students who were never awarded a voucher are identified by $selectyrs = 0$. There are 468 such students.

**(b)** Students who had a voucher available for four years are identified by $selectyrs = 4$. There are 108 such students.

**(c)** Students who attended a choice school for four years are identified by $choiceyrs = 4$. There are 56 such students.

**Table 1:** Voucher Status and Choice School Attendance

| **Statistic** | **Number of students** |
|---|---|
| Never awarded a voucher, $selectyrs = 0$ | 468 |
| Voucher available for four years, $selectyrs = 4$ | 108 |
| Attended a choice school for four years, $choiceyrs = 4$ | 56 |

## 3.2 First-Stage Regression

**(a)** The first-stage regression model is

$$choiceyrs_i = \pi_0 + \pi_1 selectyrs_i + v_i. \tag{3}$$

The OLS estimates are reported in Table 2.

**Table 2:** First-Stage Regression of $choiceyrs$ on $selectyrs$

| Variable | Coefficient | Standard Error |
|---|---|---|
| $selectyrs$ | 0.767 | 0.013 |
| Constant | 0.020 | 0.025 |
| Observations | 990 | |
| $R^2$ | 0.790 | |
| First-stage $F$-statistic | 3712.53 | |

Thus, the estimated first-stage regression is

$$\widehat{choiceyrs}_i = \underset{(0.025)}{0.020} + \underset{(0.013)}{0.767} selectyrs_i \tag{4}$$

This means that one additional year of voucher availability is associated with about 0.767 additional years of attending a choice school, on average.

**(b)** To assess the relevance of $selectyrs_i$ as an instrument for $choiceyrs_i$, we test whether $selectyrs_i$ significantly explains $choiceyrs_i$ in the first-stage regression. The hypotheses are

$$H_0 : \pi_1 = 0 \qquad \text{against} \qquad H_1 : \pi_1 \neq 0. \tag{5}$$

Under $H_0$, $selectyrs_i$ is not relevant for $choiceyrs_i$. Under $H_1$, $selectyrs_i$ is relevant for $choiceyrs_i$. The test statistic is

$$t = \frac{\hat{\pi}_1}{se(\hat{\pi}_1)} = \frac{0.7668}{0.0126} = 60.93. \tag{6}$$

Under the null hypothesis, the statistic is asymptotically standard normal:

$$t \overset{a}{\sim} N(0, 1). \tag{7}$$

Equivalently, since there is only one restriction, the corresponding first-stage $F$-statistic is

$$F = t^2 = 3712.53, \tag{8}$$

which is reported by Stata as $F(1, 988) = 3712.53$

At the 5% significance level, the two-sided critical value for the $t$-test is 1.96. The rejection rule is to reject $H_0$ if

$$|t| > 1.96. \tag{9}$$

Since $|60.93| > 1.96$, we reject $H_0$. Therefore,we have sufficient evidence that $selectyrs_i$ is strongly correlated with $choiceyrs_i$, so it satisfies the instrument relevance condition. The very large first-stage $F$-statistic also indicates that $selectyrs_i$ is not a weak instrument in this sample.

## 3.3 OLS Estimation

**(a)** The simple OLS model is

$$mnce94_i = \beta_0 + \beta_1 choiceyrs_i + u_i. \tag{10}$$

The estimated equation is

$$\widehat{mnce94}_i = \underset{(0.851)}{46.234} - \underset{(0.526)}{1.837} choiceyrs_i \tag{11}$$

**(b)** After adding $black$, $hispanic$, and $female$, the revised OLS model is

$$mnce94_i = \beta_0 + \beta_1 choiceyrs_i + \beta_2 black_i + \beta_3 hispanic_i + \beta_4 female_i + u_i. \tag{12}$$

The estimated equation is

$$\widehat{mnce94}_i = \underset{(1.657)}{57.122} - \underset{(0.531)}{0.565} choiceyrs_i - \underset{(1.794)}{16.017} black_i - \underset{(2.317)}{13.403} hispanic_i + \underset{(1.276)}{1.353} female_i \tag{13}$$

The OLS estimates are reported in Table 3.

**Table 3:** OLS Estimates for *mnce*94

| Variable | OLS Simple | OLS with Controls |
|---|---|---|
| *choiceyrs* | −1.837 | −0.565 |
| | (0.526) | (0.531) |
| *black* | | −16.017 |
| | | (1.794) |
| *hispanic* | | −13.403 |
| | | (2.317) |
| *female* | | 1.353 |
| | | (1.276) |
| Constant | 46.234 | 57.122 |
| | (0.851) | (1.657) |
| Observations | 990 | 990 |
| $R^2$ | 0.012 | 0.087 |

**(c)** From part (a) to part (b), the coefficient on *choiceyrs* changes from −1.837 to −0.565. I think part (a) may underestimate the coefficient in the sense that it gives a more negative estimate. After controlling for *black*, *hispanic*, and *female*, the estimated negative association between attending a choice school and the 1994 math percentile score becomes much smaller in absolute value.

In part (b), the coefficient on *choiceyrs* means that, holding *black*, *hispanic*, and *female* fixed, one additional year of attending a choice school is associated with a decrease of about 0.565 percentile points in *mnce*94, on average. However, this coefficient is not statistically significant at the 5% level, since its $p$-value is $0.287 > 0.05$.(We get the p-value from stata output)

Let's look at $R^2$. The $R^2$ increased from 0.012 to 0.087, which is a very small increase. This increase means that adding *black*, *hispanic*, and *female* can explain more variation in *mnce*94, although the overall explanatory power of the model remains low.

## 3.4 Why *choiceyrs* May Be Endogenous

Consider the regression model

$$mnce94_i = \beta_0 + \beta_1 choiceyrs_i + \beta_2 black_i + \beta_3 hispanic_i + \beta_4 female_i + u_i. \tag{14}$$

The variable $choiceyrs_i$ may be endogenous because the number of years a student actually attends a choice school is not randomly determined. Although voucher selection is related to the lottery, actual attendance and continuation in a choice school may depend on unobserved family and student characteristics, such as parental motivation, residential location, whether siblings or friends attend the school, school-switching costs, and prior academic ability.

These factors may affect both $choiceyrs_i$ and the student's 1994 math score $mnce94_i$. For example, Rouse (1998) notes that attendance decisions depend on factors such as family residential location and whether friends or siblings attend the school. The paper also suggests that students who remained in the choice schools may have been a self-selected group, rather than a random sample of all students who initially enrolled.

Therefore, these unobserved factors are likely to be included in the error term $u_i$, implying that

$$\text{Cov}(choiceyrs_i, u_i) \neq 0. \tag{15}$$

As a result, OLS estimates of $\beta_1$ may be biased and inconsistent. This is why lottery-based voucher selection, such as $selectyrs_i$, can be used as an instrumental variable to isolate exogenous variation in attending a choice school.

## 3.5 IV Estimation Using *selectyrs* as an Instrument

**(a)** The structural equation is

$$mnce94_i = \beta_0 + \beta_1 choiceyrs_i + \beta_2 black_i + \beta_3 hispanic_i + \beta_4 female_i + u_i. \tag{16}$$

Since $choiceyrs_i$ may be endogenous, we use $selectyrs_i$ as an instrument for $choiceyrs_i$. The corresponding first-stage equation is

$$choiceyrs_i = \pi_0 + \pi_1 selectyrs_i + \pi_2 black_i + \pi_3 hispanic_i + \pi_4 female_i + v_i. \tag{17}$$

The IV/2SLS estimated equation is

$$\begin{aligned}\widehat{mnce94}_i = \underset{(1.654)}{57.068} - \underset{(0.604)}{0.241} choiceyrs_i - \underset{(1.810)}{16.317} black_i \\ - \underset{(2.335)}{13.775} hispanic_i + \underset{(1.273)}{1.320} female_i\end{aligned} \tag{18}$$

The OLS and IV estimates are reported in Table 4. Standard errors are reported in parentheses.

Compared with the OLS estimate, the IV estimate of *choiceyrs* is less negative, changing from $-0.565$ to $-0.241$. The IV estimate is also less precise, since its standard error increases from 0.531 to 0.604. Therefore, after using lottery-based voucher selection as an instrument, the estimated association between attending a choice school and the 1994 math percentile score becomes closer to zero. If the IV assumptions are valid, the IV estimate is preferred because it uses exogenous variation in $choiceyrs_i$ generated by

**Table 4:** OLS and IV Estimates for $mnce94$

| Variable | OLS | IV/2SLS |
|---|---|---|
| $choiceyrs$ | $-0.565$ | $-0.241$ |
| | $(0.531)$ | $(0.604)$ |
| $black$ | $-16.017$ | $-16.317$ |
| | $(1.794)$ | $(1.810)$ |
| $hispanic$ | $-13.403$ | $-13.775$ |
| | $(2.317)$ | $(2.335)$ |
| $female$ | $1.353$ | $1.320$ |
| | $(1.276)$ | $(1.273)$ |
| Constant | $57.122$ | $57.068$ |
| | $(1.657)$ | $(1.654)$ |
| Observations | 990 | 990 |
| $R^2$ | 0.087 | 0.086 |

$selectyrs_i$.

**(b)** To test the significance of $choiceyrs$ in the IV regression, the hypotheses are

$$H_0 : \beta_1 = 0 \qquad \text{against} \qquad H_1 : \beta_1 \neq 0. \tag{19}$$

The test statistic is

$$z_{calc} = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} = \frac{-0.241}{0.604} = -0.40. \tag{20}$$

Under the null hypothesis,

$$z_{calc} \overset{a}{\sim} N(0, 1). \tag{21}$$

Equivalently, the Wald statistic is

$$\chi^2 = z^2 = 0.16, \qquad \chi^2 \overset{a}{\sim} \chi^2(1). \tag{22}$$

At the 5% significance level, the two-sided critical value for the $z$-test is 1.96. The rejection rule is to reject $H_0$ if

$$|z| > 1.96 \tag{23}$$

Since

$$|-0.40| < 1.96$$

we fail to reject $H_0$. Therefore, $choiceyrs$ is not statistically significant at the 5% level in the IV regression. The IV results do not provide statistically significant evidence that an additional year of attending a choice school affects the 1994 math percentile score.

## 3.6 Controlling for Prior Achievement

**(a)** To control for prior achievement, I add $mnce90_i$ as an additional regressor. The regression model is

$$mnce94_i = \beta_0 + \beta_1 choiceyrs_i + \beta_2 black_i + \beta_3 hispanic_i + \beta_4 female_i + \beta_5 mnce90_i + u_i. \quad (24)$$

For the IV estimation, $choiceyrs_i$ is treated as endogenous and $selectyrs_i$ is used as an instrument. The corresponding first-stage equation is

$$choiceyrs_i = \pi_0 + \pi_1 selectyrs_i + \pi_2 black_i + \pi_3 hispanic_i + \pi_4 female_i + \pi_5 mnce90_i + v_i. \quad (25)$$

The OLS estimated equation is

$$\begin{aligned}\widehat{mnce94}_i = \underset{(3.620)}{22.153} + \underset{(0.736)}{0.411} choiceyrs_i - \underset{(2.546)}{8.305} black_i \\ - \underset{(3.362)}{4.105} hispanic_i - \underset{(1.776)}{0.883} female_i + \underset{(0.048)}{0.620} mnce90_i\end{aligned} \quad (26)$$

The IV/2SLS estimated equation is

$$\begin{aligned}\widehat{mnce94}_i = \underset{(3.612)}{21.539} + \underset{(0.852)}{1.799} choiceyrs_i - \underset{(2.548)}{9.067} black_i \\ - \underset{(3.362)}{5.004} hispanic_i - \underset{(1.770)}{1.020} female_i + \underset{(0.048)}{0.629} mnce90_i\end{aligned} \quad (27)$$

The OLS and IV estimates are reported in Table 5.

**Table 5:** OLS and IV Estimates Controlling for Prior Achievement

| Variable | OLS with $mnce90$ | IV/2SLS with $mnce90$ |
|---|---|---|
| $choiceyrs$ | 0.411 | 1.799 |
| | (0.736) | (0.852) |
| $black$ | −8.305 | −9.067 |
| | (2.546) | (2.548) |
| $hispanic$ | −4.105 | −5.004 |
| | (3.362) | (3.362) |
| $female$ | −0.883 | −1.020 |
| | (1.776) | (1.770) |
| $mnce90$ | 0.620 | 0.629 |
| | (0.048) | (0.048) |
| Constant | 22.153 | 21.539 |
| | (3.620) | (3.612) |
| Observations | 328 | 328 |
| $R^2$ | 0.424 | 0.417 |

Standard errors are reported in parentheses.

**(b)** Before controlling for $mnce90_i$, the sample size is 990. After adding $mnce90_i$, the sample size falls to 328. Therefore, 662 observations are lost. This decrease occurs because $mnce90_i$, the prior math test score, is missing for many students, and Stata drops observations with missing values in any variable used in the regression.

**(c)** After controlling for prior achievement, the OLS estimate of the coefficient on $choiceyrs_i$ is 0.411, while the IV estimate is 1.799. Both estimates are positive, but the IV estimate is much larger than the OLS estimate.

The OLS estimate is not statistically significant at the 5% level, since its $p$-value is $0.577 > 0.05$. In contrast, the IV estimate is statistically significant at the 5% level, since its $p$-value is $0.035 < 0.05$. The IV estimate is also less precise than the OLS estimate because its standard error is larger, increasing from 0.736 to 0.852.

**(d)** For the IV estimate, the coefficient on $choiceyrs_i$ is 1.799. Therefore, holding $black_i$, $hispanic_i$, $female_i$, and $mnce90_i$ fixed, each additional year in a choice school is estimated to increase the 1994 math percentile score by about 1.799 percentile points.

Since the IV estimate is statistically significant at the 5% level, there is statistically significant IV evidence that an additional year in a choice school improves the 1994 math percentile score after controlling for prior achievement.

## 3.7 Exogeneity Test and 2SLS Using $selectyrs1$–$selectyrs4$

**(a)** To test whether $choiceyrs_i$ is exogenous in the regression equation from part 3.6, I use the control-function version of the Durbin–Wu–Hausman test. The structural equation from part 3.6 is

$$mnce94_i = \beta_0 + \beta_1 choiceyrs_i + \beta_2 black_i + \beta_3 hispanic_i + \beta_4 female_i + \beta_5 mnce90_i + u_i. \tag{28}$$

Using $selectyrs1_i, \ldots, selectyrs4_i$ as instruments, the first-stage regression is

$$\begin{aligned} choiceyrs_i = \pi_0 &+ \pi_1 selectyrs1_i + \pi_2 selectyrs2_i + \pi_3 selectyrs3_i + \pi_4 selectyrs4_i \\ &+ \pi_5 black_i + \pi_6 hispanic_i + \pi_7 female_i + \pi_8 mnce90_i + v_i. \end{aligned} \tag{29}$$

The first-stage estimates are reported in Table 6.

Standard errors are reported in parentheses. After estimating the first-stage regression, I obtain the residual

$$\hat{v}_i = choiceyrs_i - \widehat{choiceyrs}_i. \tag{30}$$

The augmented structural equation is then

$$mnce94_i = \beta_0 + \beta_1 choiceyrs_i + \beta_2 black_i + \beta_3 hispanic_i + \beta_4 female_i + \beta_5 mnce90_i + \lambda \hat{v}_i + \varepsilon_i. \tag{31}$$

The augmented regression estimates are reported in Table 7.

**Table 6:** First-Stage Regression for the Exogeneity Test

| Variable | Dependent variable: $choiceyrs$ |
|---|---|
| $selectyrs1$ | 0.945 |
| | (0.179) |
| $selectyrs2$ | 1.761 |
| | (0.142) |
| $selectyrs3$ | 1.990 |
| | (0.127) |
| $selectyrs4$ | 2.868 |
| | (0.105) |
| $black$ | −0.030 |
| | (0.099) |
| $hispanic$ | −0.128 |
| | (0.134) |
| $female$ | 0.008 |
| | (0.068) |
| $mnce90$ | −0.003 |
| | (0.002) |
| Constant | 0.178 |
| | (0.139) |
| Observations | 328 |
| $R^2$ | 0.760 |

**Table 7:** Augmented Structural Equation for the Exogeneity Test

| Variable | Dependent variable: $mnce94$ |
|---|---|
| $choiceyrs$ | 1.779 |
| | (0.839) |
| $black$ | −9.056 |
| | (2.520) |
| $hispanic$ | −4.990 |
| | (3.325) |
| $female$ | −1.018 |
| | (1.751) |
| $mnce90$ | 0.629 |
| | (0.048) |
| $\hat{v}$ | −5.411 |
| | (1.669) |
| Constant | 21.548 |
| | (3.573) |
| Observations | 328 |
| $R^2$ | 0.442 |

Standard errors are reported in parentheses. To test whether $choiceyrs_i$ is exogenous, we test the coefficient on $\hat{v}_i$. The hypotheses are

$$H_0 : \lambda = 0 \qquad \text{against} \qquad H_1 : \lambda \neq 0. \tag{32}$$

Under $H_0$, $choiceyrs_i$ is exogenous. Under $H_1$, $choiceyrs_i$ is endogenous. The test statistic is

$$t = \frac{\hat{\lambda}}{se(\hat{\lambda})} = \frac{-5.411}{1.669} = -3.24. \tag{33}$$

Under the null hypothesis,

$$t \overset{a}{\sim} N(0, 1). \tag{34}$$

Equivalently, since this is a test of one restriction,

$$F = t^2 = 10.52, \qquad F \overset{a}{\sim} \chi^2(1). \tag{35}$$

Stata reports

$$F(1, 321) = 10.52, \qquad p = 0.0013. \tag{36}$$

At the 5% significance level, the two-sided critical value for the $t$-test is 1.96. The rejection rule is to reject $H_0$ if

$$|t| > 1.96, \tag{37}$$

or equivalently if the $p$-value is less than 0.05.

Since

$$| - 3.24| > 1.96$$

and

$$0.0013 < 0.05,$$

we reject $H_0$. Therefore, there is statistically significant evidence that $choiceyrs_i$ is endogenous in the model controlling for $mnce90_i$. Hence, IV/2SLS estimation is preferred to OLS for this model.

**(b)** I next estimate model (4) using $selectyrs1_i$, $selectyrs2_i$, $selectyrs3_i$, and $selectyrs4_i$ as instruments for $choiceyrs_i$. The structural equation is

$$mnce94_i = \beta_0 + \beta_1 choiceyrs_i + \beta_2 black_i + \beta_3 hispanic_i + \beta_4 female_i + u_i. \tag{38}$$

Here, $choiceyrs_i$ is treated as endogenous, and the four *selectyrs* dummy variables are used as excluded instruments. The 2SLS estimates are reported in Table 8.

Standard errors are reported in parentheses. The coefficient on $choiceyrs_i$ is $-0.253$, with standard error 0.603. This estimate is close to zero and is not statistically significant at the 5% level, since its $p$-value is $0.675 > 0.05$.

**Table 8:** 2SLS Estimates Using *selectyrs*1–*selectyrs*4 as Instruments

| Variable | IV/2SLS |
|---|---|
| $choiceyrs$ | $-0.253$ |
| | (0.603) |
| $black$ | $-16.306$ |
| | (1.810) |
| $hispanic$ | $-13.762$ |
| | (2.335) |
| $female$ | 1.321 |
| | (1.273) |
| Constant | 57.070 |
| | (1.653) |
| Observations | 990 |
| $R^2$ | 0.087 |

## 3.8 Overidentification Test

Since there are four excluded instruments, $selectyrs1_i$, $selectyrs2_i$, $selectyrs3_i$, and $selectyrs4_i$, for one endogenous variable, $choiceyrs_i$, the model is overidentified. The number of overidentifying restrictions is

$$q = 4 - 1 = 3. \tag{39}$$

To test the joint exogeneity of the four *selectyrs* instruments, I use an overidentification test based on the IV residuals. Using the 2SLS model reported in Table 8, let $\hat{u}_i$ denote the IV residuals from

$$mnce94_i = \beta_0 + \beta_1 choiceyrs_i + \beta_2 black_i + \beta_3 hispanic_i + \beta_4 female_i + u_i, \tag{40}$$

where $choiceyrs_i$ is instrumented by $selectyrs1_i, \ldots, selectyrs4_i$.
I then estimate the auxiliary regression

$$\begin{aligned}\hat{u}_i = {} & \alpha_0 + \alpha_1 black_i + \alpha_2 hispanic_i + \alpha_3 female_i \\ & + \alpha_4 selectyrs1_i + \alpha_5 selectyrs2_i + \alpha_6 selectyrs3_i + \alpha_7 selectyrs4_i + e_i.\end{aligned} \tag{41}$$

The hypotheses are

$$H_0 : \mathrm{Cov}(selectyrsj_i, u_i) = 0 \quad \text{for all } j = 1, 2, 3, 4, \tag{42}$$

against

$$H_1 : \mathrm{Cov}(selectyrsj_i, u_i) \neq 0 \quad \text{for at least one } j. \tag{43}$$

Equivalently, $H_0$ states that all overidentifying restrictions are valid, while $H_1$ states that at least one instrument is not exogenous. The test statistic is

$$LM = nR^2, \tag{44}$$

where $R^2$ is obtained from the auxiliary regression. Under $H_0$,

$$LM \overset{a}{\sim} \chi^2(q) = \chi^2(3). \tag{45}$$

From the auxiliary regression, $n = 990$ and $R^2 = 0.0016$. Using the unrounded value of $R^2$, the test statistic is

$$LM = nR^2 = 1.571. \tag{46}$$

The overidentification test results are reported in Table 9.

**Table 9:** Overidentification Test for the Four *selectyrs* Instruments

| Test component | Value |
|---|---|
| Number of observations, $n$ | 990 |
| Auxiliary regression $R^2$ | 0.0016 |
| Test statistic, $LM = nR^2$ | 1.571 |
| Degrees of freedom, $q = 4 - 1$ | 3 |
| 5% critical value, $\chi^2_{0.95,3}$ | 7.815 |
| $p$-value | 0.666 |

At the 5% significance level, the rejection rule is to reject $H_0$ if

$$LM > \chi^2_{0.95,3} = 7.815, \tag{47}$$

or equivalently if the $p$-value is less than 0.05.
Since

$$1.571 < 7.815$$

and

$$0.666 > 0.05,$$

we fail to reject $H_0$. Therefore, there is no statistically significant evidence against the joint exogeneity of the four *selectyrs* instruments. The overidentification test does not reject the validity of the overidentifying restrictions.

# Question 4: Causal Inference

## (i)

Let $D_i$ denote the treatment status for patient $i$, where

$$D_i = \begin{cases} 1, & \text{if patient } i \text{ received RHC on day 1,} \\ 0, & \text{if patient } i \text{ did not receive RHC on day 1.} \end{cases}$$

In the potential outcomes framework, $Y_i(1)$ is the potential outcome for patient $i$ if the patient received RHC on day 1, and $Y_i(0)$ is the potential outcome for patient $i$ if the patient did not receive RHC on day 1. In this example, the outcome is whether the patient died within 180 days, so

$$Y_i(1) = \text{death status within 180 days if patient } i \text{ received RHC},$$

and

$$Y_i(0) = \text{death status within 180 days if patient } i \text{ did not receive RHC}.$$

The observed outcome is related to the potential outcomes by

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0). \tag{48}$$

Equivalently,

$$Y_i = \begin{cases} Y_i(1), & \text{if } D_i = 1, \\ Y_i(0), & \text{if } D_i = 0. \end{cases}$$

## (ii)

The sample contains 5735 observations. Since $rhc_i$ is a dummy variable equal to one if patient $i$ received RHC on day 1, the fraction of patients who received the RHC treatment is the sample mean of $rhc_i$:

$$\overline{rhc} = \frac{1}{n} \sum_{i=1}^{n} rhc_i = \frac{2184}{5735} = 0.381. \tag{49}$$

Therefore, 2184 patients received the RHC treatment, which is approximately 38.08% of the sample.

## (iii)

To estimate the simple difference in mean death rates between patients who received RHC and those who did not, I estimate the following linear regression with robust standard errors:

$$death_i = \alpha + \tau rhc_i + u_i. \tag{50}$$

Since $rhc_i$ is a dummy variable equal to one if patient $i$ received RHC, the coefficient $\tau$ measures the difference in mean death rates between the treated and untreated groups:

$$\tau = E(death_i \mid rhc_i = 1) - E(death_i \mid rhc_i = 0). \tag{51}$$

From the Stata output, we can get the regression model:

$$\widehat{death}_i = \underset{(0.008)}{0.630} + \underset{(0.013)}{0.051} rhc_i. \tag{52}$$

The results are reported in Table 10.

**Table 10:** Simple Difference in Mean Death Rates

| Variable | Dependent variable: *death* |
|---|---|
| $rhc$ | 0.051 |
| | (0.013) |
| Constant | 0.630 |
| | (0.008) |
| Observations | 5735 |
| $R^2$ | 0.003 |

Robust standard errors are reported in parentheses. The estimated coefficient on $rhc_i$ is 0.051, with a robust standard error of 0.013. This means that patients who received RHC had a death rate about 5.1 percentage points higher than patients who did not receive RHC. The estimated death rate for non-RHC patients is 0.630, while the estimated death rate for RHC patients is

$$0.630 + 0.051 = 0.681.$$

The difference is statistically significant at the 5% level, since the robust $t$-statistic is 3.95 and the $p$-value is less than 0.05.

However, this simple difference in means is not necessarily an unbiased estimator of the ATT. In potential-outcomes notation,

$$\begin{aligned} E(Y_i \mid D_i = 1) - E(Y_i \mid D_i = 0) &= E(Y_i(1) \mid D_i = 1) - E(Y_i(0) \mid D_i = 0) \\ &= \underbrace{E(Y_i(1) - Y_i(0) \mid D_i = 1)}_{\text{ATT}} + \underbrace{[E(Y_i(0) \mid D_i = 1) - E(Y_i(0) \mid D_i = 0)]}_{\text{selection bias}}. \end{aligned} \tag{53}$$

Therefore, the simple difference in mean death rates is unbiased for the ATT only if

$$E(Y_i(0) \mid D_i = 1) = E(Y_i(0) \mid D_i = 0).$$

This condition is unlikely to hold here because RHC treatment was not randomly assigned. Patients who received RHC were likely to differ systematically from those who did not, for example in illness severity and other health characteristics. Therefore, the estimated difference may reflect both the treatment effect and selection bias, rather than the causal ATT alone.

## (iv)

I use separate linear regression adjustment (SRA) to estimate the ATT. Let

$$X_i = (age_i, sex_i, race_i, income_i, cat1_i, cat2_i, ninsclas_i)$$

denote the set of covariates. The separate outcome regressions are

$$E(death_i \mid rhc_i = 0, X_i) = \alpha_0 + X_i'\gamma_0, \tag{54}$$
$$E(death_i \mid rhc_i = 1, X_i) = \alpha_1 + X_i'\gamma_1. \tag{55}$$

The results are reported in Table 11.

The SRA estimate of the ATT is 0.0642, with a robust standard error of 0.0132. This means that, after controlling for age, sex, race, income, primary disease category, secondary disease category, and medical insurance category, patients who received RHC had an estimated death probability about 6.42 percentage points higher than comparable patients who did not receive RHC.

This estimate is slightly larger than the simple difference in mean death rates from part (iii), which was 0.051, or 5.1 percentage points. The difference occurs because SRA adjusts for observed covariates, while the simple difference in means does not.

The SRA estimate will be consistent for the true ATT if treatment assignment is conditionally unconfounded given the observed covariates, so that

$$Y_i(0) \perp rhc_i \mid X_i,$$

and if there is sufficient overlap between treated and untreated patients. Since SRA is based on linear outcome regressions, the outcome models also need to be correctly specified.

**Table 11:** Separate Linear Regression Adjustment Estimates

| Variable | ATET | POmean | OME0 | OME1 |
|---|---|---|---|---|
| age | | | 0.00681*** | 0.00533*** |
| | | | (0.000581) | (0.000788) |
| 2.sex | | | 0.0495*** | 0.00162 |
| | | | (0.0152) | (0.0194) |
| 2.race | | | 0.0332 | 0.0355 |
| | | | (0.0350) | (0.0447) |
| 3.race | | | -0.00861 | -0.0104 |
| | | | (0.0213) | (0.0271) |
| 2.income | | | -0.0251 | -0.0246 |
| | | | (0.0274) | (0.0326) |
| 3.income | | | -0.0272 | -0.0185 |
| | | | (0.0338) | (0.0400) |
| 4.income | | | 0.0333* | 0.0629** |
| | | | (0.0198) | (0.0252) |
| 2.cat1 | | | 0.0295 | 0.0746** |
| | | | (0.0342) | (0.0371) |
| 3.cat1 | | | 0.0168 | 0.0979* |
| | | | (0.0270) | (0.0588) |
| 4.cat1 | | | 0.223*** | 0.279*** |
| | | | (0.0373) | (0.0543) |
| 5.cat1 | | | 0.0103 | -0.569*** |
| | | | (0.212) | (0.0533) |
| 6.cat1 | | | 0.234*** | 0.252*** |
| | | | (0.0254) | (0.0371) |
| 7.cat1 | | | 0.355*** | 0.382*** |
| | | | (0.0532) | (0.0310) |
| 8.cat1 | | | 0.389*** | 0.349*** |
| | | | (0.0249) | (0.0284) |
| 9.cat1 | | | 0.108*** | 0.0995*** |
| | | | (0.0243) | (0.0254) |
| 2.cat2 | | | 0.0845 | -0.0255 |
| | | | (0.0774) | (0.0325) |
| 3.cat2 | | | -0.0983 | -0.301*** |
| | | | (0.0882) | (0.0901) |
| 4.cat2 | | | 0.107 | -0.159*** |
| | | | (0.0771) | (0.0387) |
| 5.cat2 | | | 0.0399 | -0.0916** |
| | | | (0.0786) | (0.0424) |
| 6.cat2 | | | -0.235*** | -0.450*** |
| | | | (0.0776) | (0.0332) |
| 7.cat2 | | | -0.356*** | -0.495*** |
| | | | (0.0753) | (0.0310) |
| 2.ninsclas | | | 0.0487 | 0.0820* |
| | | | (0.0307) | (0.0423) |
| 3.ninsclas | | | 0.0924** | 0.0344 |
| | | | (0.0373) | (0.0544) |
| 4.ninsclas | | | 0.0454 | -0.0286 |
| | | | (0.0389) | (0.0515) |
| 5.ninsclas | | | 0.0279 | 0.00490 |
| | | | (0.0284) | (0.0395) |
| 6.ninsclas | | | 0.0308 | 0.0267 |
| | | | (0.0317) | (0.0435) |
| 1.*rhc* vs 0.*rhc* | 0.0642*** | | | |
| | (0.0132) | | | |
| 0.*rhc* | | 0.616*** | | |
| | | (0.00994) | | |
| Constant | | | 0.367*** | 0.693*** |
| | | | (0.0835) | (0.0543) |
| Observations | 5,735 | 5,735 | 5,735 | 5,735 |

Robust standard errors are reported in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

**v**

I estimate the conditional probability of receiving RHC using a logit model with the same covariates as in part (iv). Let $D_i = rhc_i$. The propensity score is

$$p(X_i) = P(D_i = 1 \mid X_i), \tag{56}$$

where $X_i$ includes $age_i$, $sex_i$, $race_i$, $income_i$, $cat1_i$, $cat2_i$, and $ninsclas_i$.
The logit model is

$$P(rhc_i = 1 \mid X_i) = \Lambda(X_i'\gamma), \qquad \Lambda(z) = \frac{\exp(z)}{1 + \exp(z)}. \tag{57}$$

The logit regression results are reported in Table 12.

**Table 12:** Logit Model for Receiving RHC

| Variable | Coefficient | Robust SE |
|---|---|---|
| age | 0.000642 | (0.00222) |
| 2.sex | 0.163*** | (0.0588) |
| 2.race | 0.0394 | (0.136) |
| 3.race | 0.0424 | (0.0828) |
| 2.income | 0.107 | (0.0984) |
| 3.income | 0.114 | (0.121) |
| 4.income | -0.0444 | (0.0763) |
| 2.cat1 | 0.498*** | (0.107) |
| 3.cat1 | -1.226*** | (0.150) |
| 4.cat1 | -0.717*** | (0.171) |
| 5.cat1 | -1.003 | (1.085) |
| 6.cat1 | -0.694*** | (0.126) |
| 7.cat1 | -1.259*** | (0.483) |
| 8.cat1 | -0.208* | (0.118) |
| 9.cat1 | 1.004*** | (0.0768) |
| 2.cat2 | 0.980 | (1.465) |
| 3.cat2 | -0.414 | (0.443) |
| 4.cat2 | -0.886 | (0.845) |
| 5.cat2 | -0.195 | (0.393) |
| 6.cat2 | 1.034*** | (0.370) |
| 7.cat2 | 0.142 | (0.365) |
| 2.ninsclas | 0.185 | (0.122) |
| 3.ninsclas | 0.108 | (0.152) |
| 4.ninsclas | 0.522*** | (0.150) |
| 5.ninsclas | 0.468*** | (0.112) |
| 6.ninsclas | 0.374*** | (0.125) |
| Constant | -1.322*** | (0.389) |
| Observations | 5,735 | |

Robust standard errors are reported in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The logit model uses 5,735 observations. The estimated propensity scores are summarised in Table 13.

**Table 13:** Summary of Estimated Propensity Scores

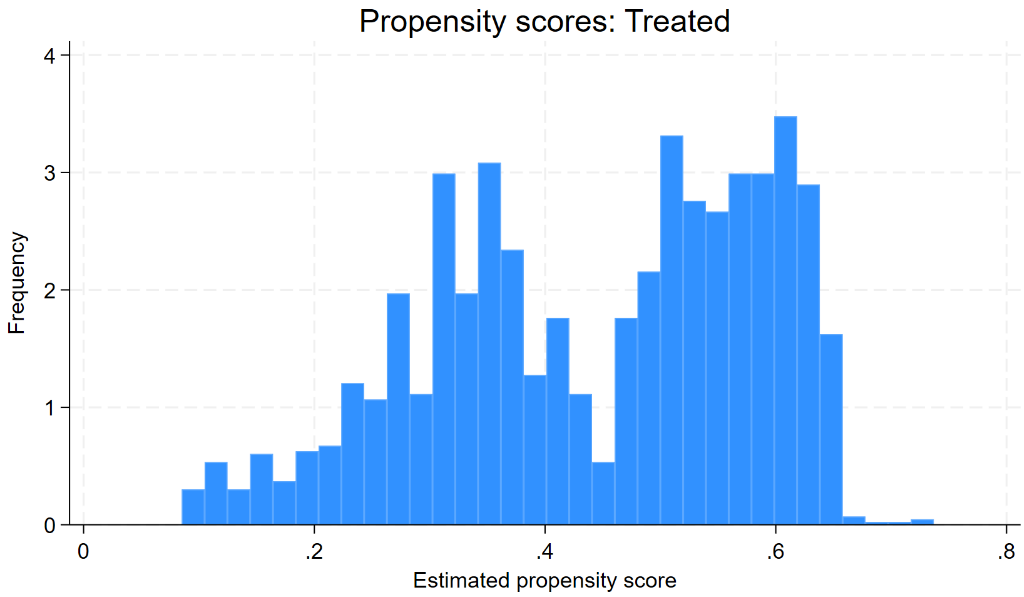
| Group | Mean | Min | Max | Range | Observations |
|---|---|---|---|---|---|
| Untreated, $rhc = 0$ | 0.3414 | 0.0447 | 0.7380 | 0.6933 | 3551 |
| Treated, $rhc = 1$ | 0.4450 | 0.0851 | 0.7369 | 0.6518 | 2184 |
| Total | 0.3809 | 0.0447 | 0.7380 | 0.6933 | 5735 |

The treated group has a higher average estimated propensity score than the untreated group. The mean propensity score is 0.4450 for treated patients, compared with 0.3414 for untreated patients. This suggests that, based on observed covariates, patients who received RHC were more likely to have characteristics associated with receiving the treatment.

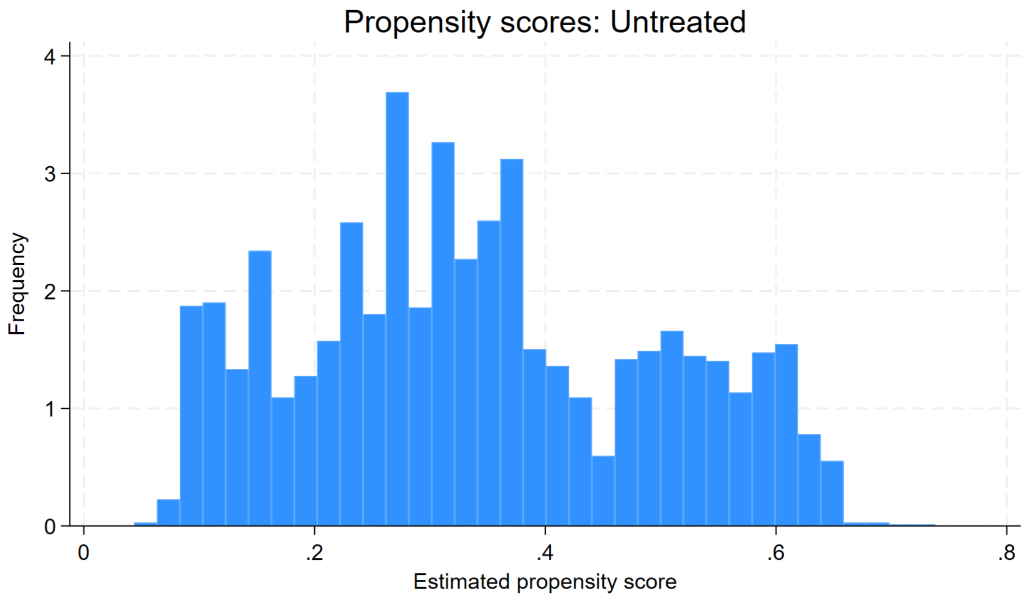
The ranges of the estimated propensity scores show substantial overlap between the two groups. For treated patients, the estimated propensity scores range from 0.0851 to 0.7369. For untreated patients, they range from 0.0447 to 0.7380. Therefore, although the treated group has a higher average propensity score, the two distributions have a large common support region.

The histograms in Figure 1 also show overlap in the estimated propensity score distributions for the treated and untreated groups. This supports the overlap condition, although treatment assignment is clearly related to observed patient characteristics.

Overall, the propensity score results indicate that treatment assignment is related to observed covariates, since the average propensity score is higher for treated patients. However, the overlapping ranges suggest that the common support condition is reasonably plausible for much of the sample.




Treated group, $rhc = 1$




Untreated group, $rhc = 0$

**Figure 1:** Histograms of Estimated Propensity Scores by Treatment Status

## vi

I estimate the ATT using propensity score weighting, where the propensity score is estimated using a logit model with the same covariates as in part 4.5. Let

$$p(X_i) = P(D_i = 1 \mid X_i),$$

where $D_i = rhc_i$, and $X_i$ includes $age_i$, $sex_i$, $race_i$, $income_i$, $cat1_i$, $cat2_i$, and $ninsclas_i$. The treatment model is

$$P(rhc_i = 1 \mid X_i) = \Lambda(X_i'\gamma), \qquad \Lambda(z) = \frac{\exp(z)}{1 + \exp(z)}. \tag{58}$$

For the ATT, propensity score weighting reweights the untreated group so that its observed covariate distribution resembles that of the treated group. The results are reported in Table 14.

**Table 14:** Propensity Score Weighting Estimate of the ATT

| Parameter | Estimate |
|---|---|
| ATT, $rhc : 1$ vs 0 | 0.064 |
| | (0.013) |
| Predicted mean outcome under no RHC for treated group | 0.617 |
| | (0.010) |
| Observations | 5735 |
| Treatment model | Logit |

Robust standard errors are reported in parentheses. The propensity score weighting estimate of the ATT is 0.064, with a robust standard error of 0.013. This means that, after reweighting the untreated group using the estimated propensity scores, receiving RHC is estimated to increase the probability of dying within 180 days by about 6.4 percentage points for the treated patients.

This estimate is very close to the SRA estimate from part 4.4, which was 0.064. Therefore, both regression adjustment and propensity score weighting give a similar estimate of the ATT. Both estimates are larger than the simple difference in means from part 4.3, which was 0.051. This suggests that adjusting for observed covariates increases the estimated effect of RHC on 180-day mortality.

The PSW estimator is consistent for the true ATT if the conditional independence assumption holds for the untreated potential outcome,

$$Y_i(0) \perp D_i \mid X_i, \tag{59}$$

so that, conditional on the observed covariates, untreated patients provide a valid comparison group for treated patients. It also requires overlap or common support:

$$0 < P(D_i = 1 \mid X_i) < 1. \tag{60}$$

From part 4(v), the estimated propensity score ranges overlap substantially between the treated and untreated groups, which supports the common support condition. Finally, the logit propensity score model must be correctly specified. If important confounders are omitted, or if there is poor overlap in propensity scores, the PSW estimate may still be biased.